

Question 1

```
In [7]: import matplotlib.pyplot as plt
import numpy as np

values = {
    'A': 13.85, 'D': 11.61, 'C': 15.37, 'E': 11.38, 'F': 13.93, 'G': 13.34,
    'H': 13.82, 'I': 15.28, 'K': 11.58, 'L': 14.13, 'M': 13.86, 'N': 13.02,
    'P': 12.35, 'Q': 12.61, 'R': 13.10, 'S': 13.39, 'T': 12.70, 'V': 14.56,
    'W': 15.48, 'Y': 13.88
}

f = open("q1.fasta", mode="r")
sequences = [seq.strip() for seq in f.readlines()[1::2]]
f.close()

fig, ax = plt.subplots(nrows=3, dpi=200, figsize=(10, 10))

count = 0
for sequence in sequences:
    hydrophobicity_profile = [values[residue] for residue in sequence]
    average = np.mean(hydrophobicity_profile)
    deviation = [0 if hydrophobicity < average else 1 for hydrophobicity in hydrophobicity_profile]
    helices = set()
    sheets = set()

    for i in range(len(sequence) - 4 + 1):
        if deviation[i] == deviation[i + 1] and deviation[i] != deviation[i + 2] and deviation[i + 2] == deviation[i + 3]:
            helices.update({i, i + 1, i + 2, i + 3})
        elif deviation[i] != deviation[i + 1] and deviation[i] == deviation[i + 2] and deviation[i] != deviation[i + 3]:
            sheets.update({i, i + 1, i + 2, i + 3})

    ax[count].plot(range(len(sequence)), hydrophobicity_profile, c="black", marker="o", markersize=3)
    ax[count].set_title(f"\nHydrophobicity profile of sequence {count + 1}")
    ax[count].set_xlabel("Residue number")
    ax[count].set_ylabel("Hydrophobicity value")
    ax[count].axhline(average, c="black", ls="--", lw=0.75)

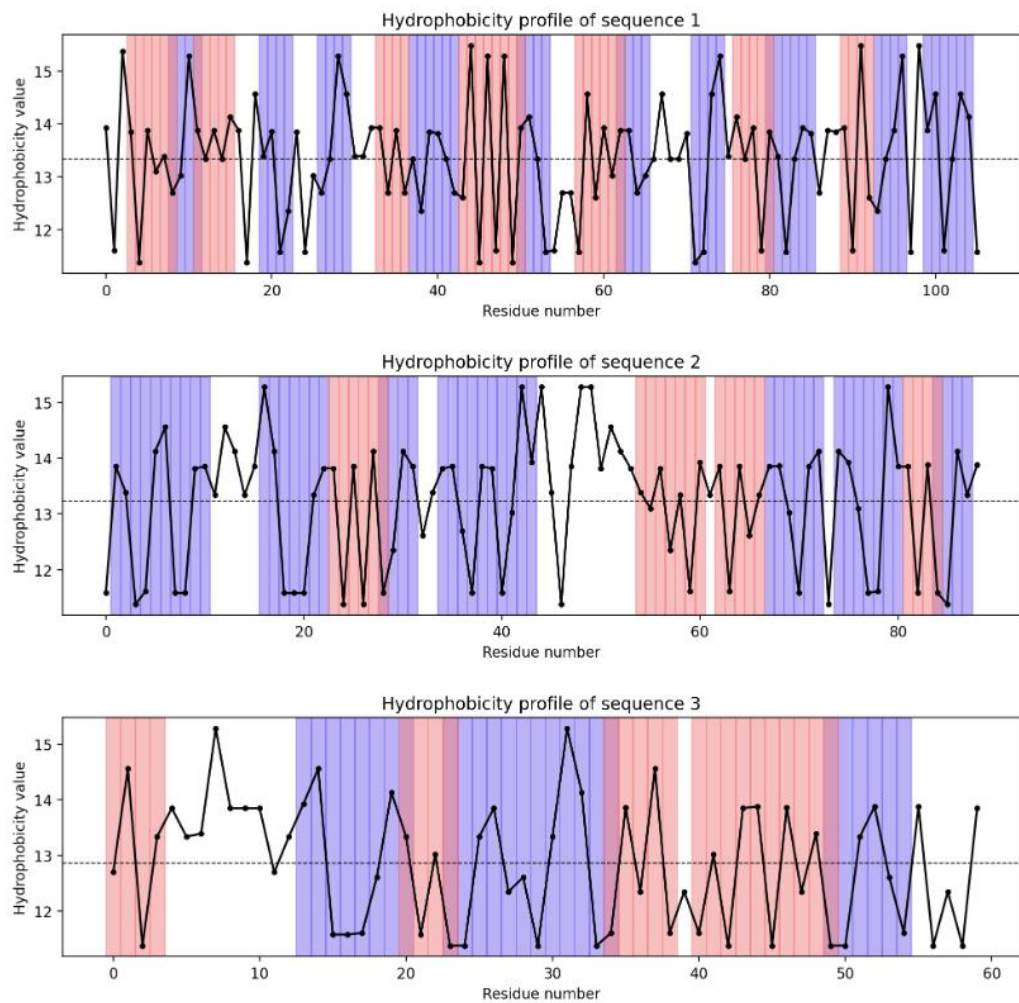
    for position in helices:
        ax[count].axvspan(position - 0.5, position + 0.5, alpha=0.5, color='mediumslateblue')

    for position in sheets:
        ax[count].axvspan(position - 0.5, position + 0.5, alpha=0.5, color='lightcoral')

    count += 1

fig.tight_layout()
plt.show()
```

The output of this, with α -helices marked in blue and β -strands in red, is shown below:



Question 2

```
In [8]: values = {
        'A': 13.85, 'D': 11.61, 'C': 15.37, 'E': 11.38, 'F': 13.93, 'G': 13.34,
        'H': 13.82, 'I': 15.28, 'K': 11.58, 'L': 14.13, 'M': 13.86, 'N': 13.02,
        'P': 12.35, 'Q': 12.61, 'R': 13.10, 'S': 13.39, 'T': 12.70, 'V': 14.56,
        'W': 15.48, 'Y': 13.88
    }

    f = open("Q1.fasta", mode="r")
    sequences = [seq.strip() for seq in f.readlines()[1::2]]
    f.close()

    count = 1
    for sequence in sequences:
        hydrophobicity_profile = [values[residue] for residue in sequence]
        average = np.mean(hydrophobicity_profile)
        deviation = [0 if hydrophobicity < average else 1 for hydrophobicity in hydrophobicity_profile]
        alphas = [i for i in range(len(sequence) - 8 + 1) if ((deviation[i:i + 8] == [0, 0, 1, 1, 0, 0, 1, 1]) or (deviation[i:i + 8] == [1, 1, 0, 0, 1, 1, 0, 0]))]
        betas = [i for i in range(len(sequence) - 6 + 1) if ((deviation[i:i + 6] == [0, 1, 0, 1, 0, 1]) or (deviation[i:i + 6] == [1, 0, 1, 0, 1, 0]))]
        alpha_amphipathicities = []
        for alpha_start in alphas:
            a1 = 0
            a2 = 0
            a3 = 0
            a4 = 0
            for j in range(0, 8, 4):
                a1 += values[sequence[alpha_start + j]]
                a2 += values[sequence[alpha_start + 1 + j]]
                a3 += values[sequence[alpha_start + 2 + j]]
                a4 += values[sequence[alpha_start + 3 + j]]
            a1 = a1 / 2
            a2 = a2 / 2
            a3 = a3 / 2
            a4 = a4 / 2
            if deviation[alpha_start] == deviation[alpha_start + 1]:
                alpha_amphipathicities.append(abs((a1 + a2) - (a3 + a4)))
            else:
                alpha_amphipathicities.append(abs((a1 + a4) - (a2 + a3)))
        beta_amphipathicities = []
        for beta_start in betas:
            b1 = 0
            b2 = 0
            for j in range(0, 6, 2):
                b1 += values[sequence[beta_start + j]]
                b2 += values[sequence[beta_start + 1 + j]]
            b1 = b1 / 3
            b2 = b2 / 3
            if deviation[beta_start] == 0:
                beta_amphipathicities.append(b2 - b1)
            else:
                beta_amphipathicities.append(b1 - b2)
        print(f"Sequence {count}")
        if len(alphas) != 0:
            for i in range(len(alphas)):
                print(f"Helix of length 8 found at position {alphas[i]} with amphipathicity {round(alpha_amphipathicities[i], 3)}")
        else:
            print("No helices of length 8 were found")
        if len(betas) != 0:
            for i in range(len(betas)):
                print(f"Sheet of length 6 found at position {betas[i]} with amphipathicity {round(beta_amphipathicities[i], 3)}")
        else:
            print("No sheets of length 6 were found")
        print("")
        count += 1
```

```
Sequence 1
No helices of length 8 were found
Sheet of length 6 found at position 3 with amphipathicity 1.313
Sheet of length 6 found at position 43 with amphipathicity 3.48
Sheet of length 6 found at position 44 with amphipathicity 3.89
Sheet of length 6 found at position 45 with amphipathicity 3.373
Sheet of length 6 found at position 57 with amphipathicity 1.72
```

```
Sequence 2
Helix of length 8 found at position 1 with amphipathicity 4.89
Helix of length 8 found at position 3 with amphipathicity 5.105
Helix of length 8 found at position 34 with amphipathicity 3.23
Helix of length 8 found at position 36 with amphipathicity 4.0
Sheet of length 6 found at position 23 with amphipathicity 2.487
Sheet of length 6 found at position 54 with amphipathicity 1.163
Sheet of length 6 found at position 55 with amphipathicity 1.343
```

```
Sequence 3
No helices of length 8 were found
Sheet of length 6 found at position 44 with amphipathicity 2.007
```

Question 3

```
In [12]: values = {
    'A': 13.85, 'D': 11.61, 'C': 15.37, 'E': 11.38, 'F': 13.93, 'G': 13.34,
    'H': 13.82, 'I': 15.28, 'K': 11.58, 'L': 14.13, 'M': 13.86, 'N': 13.02,
    'P': 12.35, 'Q': 12.61, 'R': 13.10, 'S': 13.39, 'T': 12.70, 'V': 14.56,
    'W': 15.48, 'Y': 13.88
}

f = open("Q2.fasta", mode="r")
sequences = [seq.strip() for seq in f.readlines()[1::2]]
f.close()

window_size1 = 9
half_width1 = window_size1 // 2
window_size2 = 19
half_width2 = window_size2 // 2

for sequence in sequences:
    hydrophobicity_profile = [np.mean([values[residue] for residue in sequence[i - half_width1:i + half_width1 + 1]]) for i in range(len(sequence) - half_width1)]
    average = np.mean(hydrophobicity_profile)
    deviation = [0 if hydrophobicity < average else 1 for hydrophobicity in hydrophobicity_profile]
    transmembrane = set()

    for i in range(len(hydrophobicity_profile) - 4 + 1):
        if deviation[i] == 1 and deviation[i] == deviation[i + 1] and deviation[i + 1] == deviation[i + 2] and deviation[i + 2] == deviation[i + 3]:
            transmembrane.update({i, i + 1, i + 2, i + 3})

    plt.figure(dpi=200, figsize=(10, 5))
    plt.plot(range(len(hydrophobicity_profile)), hydrophobicity_profile, c="black", marker="o", markersize=3)
    plt.title(f"Hydrophobicity profile of sequence with window length {window_size1}")
    plt.xlabel("Residue number")
    plt.ylabel("Hydrophobicity value")
    plt.axhline(average, c="black", ls="--", lw=0.75)

    for position in transmembrane:
        plt.axvspan(position - 0.5, position + 0.5, alpha=0.5, color='yellow')

    plt.show()

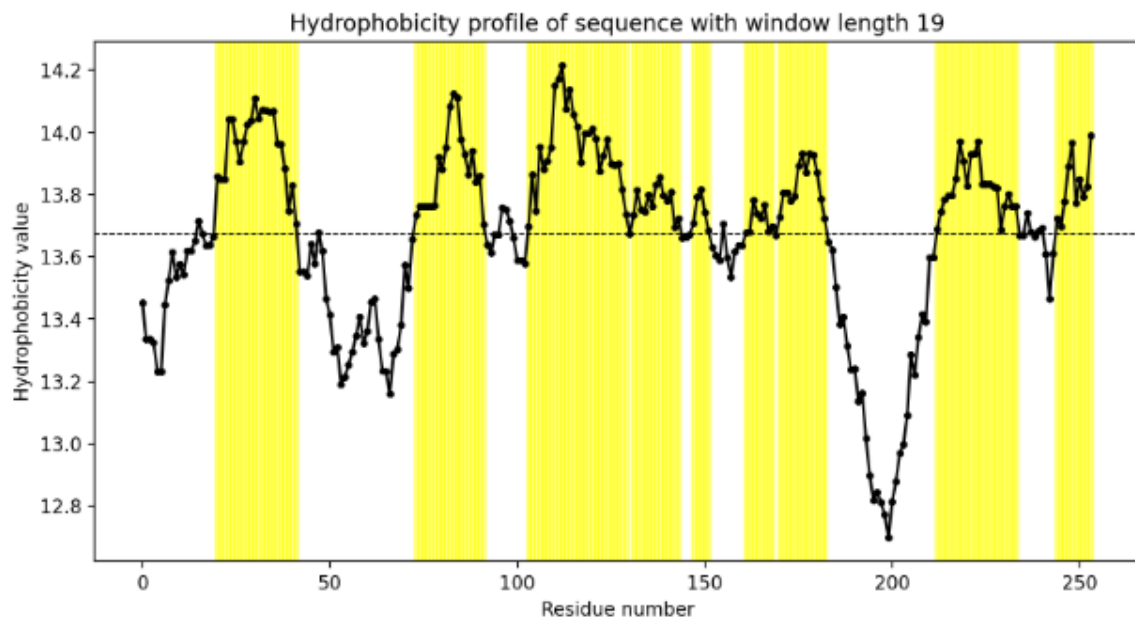
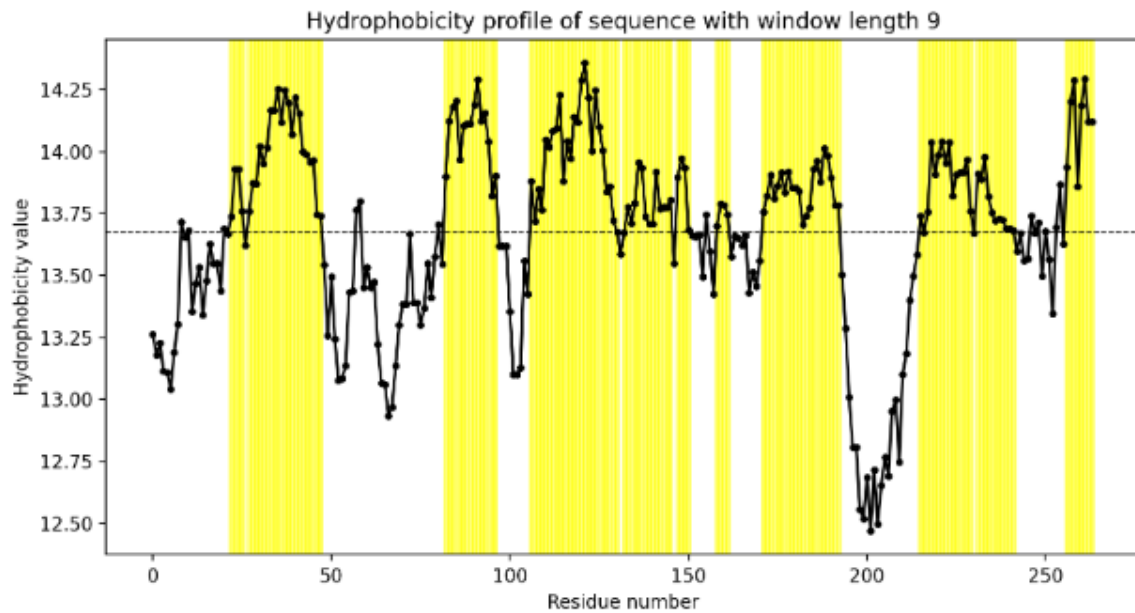
    hydrophobicity_profile = [np.mean([values[residue] for residue in sequence[i - half_width2:i + half_width2 + 1]]) for i in range(len(sequence) - half_width2)]
    average = np.mean(hydrophobicity_profile)
    deviation = [0 if hydrophobicity < average else 1 for hydrophobicity in hydrophobicity_profile]
    transmembrane = set()

    for i in range(len(hydrophobicity_profile) - 4 + 1):
        if deviation[i] == 1 and deviation[i] == deviation[i + 1] and deviation[i + 1] == deviation[i + 2] and deviation[i + 2] == deviation[i + 3]:
            transmembrane.update({i, i + 1, i + 2, i + 3})

    plt.figure(dpi=200, figsize=(10, 5))
    plt.plot(range(len(hydrophobicity_profile)), hydrophobicity_profile, c="black", marker="o", markersize=3)
    plt.title(f"Hydrophobicity profile of sequence with window length {window_size2}")
    plt.xlabel("Residue number")
    plt.ylabel("Hydrophobicity value")
    plt.axhline(average, c="black", ls="--", lw=0.75)

    for position in transmembrane:
        plt.axvspan(position - 0.5, position + 0.5, alpha=0.5, color='yellow')

    plt.show()
```



ScanProsite tool

This form requires to have JavaScript enabled to work correctly.

This form allows you to scan proteins for matches against the [PROSITE collection of motifs](#) as well as against your own patterns.

- ☐ Option 1 - Submit PROTEIN sequences to scan them against the PROSITE collection of motifs.
- ☒ **Option 2 - Submit MOTIFS to scan them against a PROTEIN sequence database.**
- ☐ Option 3 - Submit PROTEIN sequences and MOTIFS to scan them against each other.

[Reset](#)

STEP 1 - Enter a MOTIF or a combination of MOTIFS [Examples](#) [help](#)

[SV]-T-[VT]-[DERK](2)-[IL]



Supported input:

Question 4

For the pattern [SV]-T-[VT]-[DERK](2)-{IL}, 11007 hits were found in 10584 sequences.

[ScanProsite] Results for job: Pattern A External Inbox x



Expasy <expasy@expasy.org>

to me ▾

ScanProsite results:

WARNING: result output bigger than 4MB!
output truncated to 4MB
try to change your scan search parameters to get less results!

include splice variants (UniProtKB/Swiss-Prot)
Output format: Text

Hits for USERPAT1 "[SV]-T-[VT]-[DERK](2)-{IL}" on UniProtKB/Swiss-Prot sequences:
UniProtKB/Swiss-Prot (Release 2024_02 of 27-Mar-24) contains 571'282 entries.

found: 11007 hits in 10584 sequences

For the pattern [FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY], 3837 hits were found in 3774 sequences.



Expasy <expasy@expasy.org>

to me ▾

ScanProsite results:

WARNING: result output bigger than 4MB!
output truncated to 4MB
try to change your scan search parameters to get less results!

include splice variants (UniProtKB/Swiss-Prot)
Output format: Text

Hits for USERPAT1 "[FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY]" on UniProtKB/Swiss-Prot sequences:
UniProtKB/Swiss-Prot (Release 2024_02 of 27-Mar-24) contains 571'282 entries.

found: 3837 hits in 3774 sequences

Question 5

```
In [14]: f = open("Q4.fasta", mode="r")
content = f.readlines()
sequences = {content[i][1:].strip(): content[i + 1].strip() for i in range(0, len(content) - 1, 2)}
f.close()

for header, sequence in sequences.items():
    pattern1_length = 6
    pattern2_length = 14

    for i in range(len(sequence) - pattern1_length + 1):
        sub_sequence = sequence[i:i + pattern1_length]
        if (sub_sequence[0] == "S" or sub_sequence[0] == "V") and (sub_sequence[1] == "T") and \
            (sub_sequence[2] == "V" or sub_sequence[2] == "T") and \
            (sub_sequence[3] == "D" or sub_sequence[3] == "E" or sub_sequence[3] == "R" or sub_sequence[3] == "K") and \
            (sub_sequence[4] == "D" or sub_sequence[4] == "E" or sub_sequence[4] == "R" or sub_sequence[4] == "K") and \
            (sub_sequence[5] != "I" and sub_sequence[5] != "L"):
            print(f"Match for Pattern 1 found in {header} at position {i + 1}")

    for i in range(len(sequence) - pattern2_length + 1):
        sub_sequence = sequence[i:i + pattern2_length]
        if (sub_sequence[0] in ["F", "I", "L", "V"]) and (sub_sequence[1] == "Q") and \
            (sub_sequence[5] not in ["R", "K"]) and (sub_sequence[6] == "G") and \
            (sub_sequence[10] in ["R", "K"]) and (sub_sequence[13] in ["F", "I", "L", "V", "W", "Y"]):
            print(f"Match for Pattern 2 found in {header} at position {i + 1}")
```

```
Match for Pattern 1 found in 4A0C_2|Chains C,E|CULLIN-4B|HOMO SAPIENS (9606) at position 665
Match for Pattern 1 found in 4A0K_1|Chain A|CULLIN-4A|HOMO SAPIENS (9606) at position 666
Match for Pattern 2 found in 4FXG_2|Chains B,E|Complement C4-A alpha chain|Homo sapiens (9606) at position 252
Match for Pattern 2 found in 4FXK_2|Chain B|Complement C4-A Alpha chain|Homo sapiens (9606) at position 252
Match for Pattern 2 found in 4XAM_2|Chains C,E|Complement C4-A|Homo sapiens (9606) at position 175
Match for Pattern 1 found in 5F0J_3|Chain C|Sorting nexin-3|Homo sapiens (9606) at position 70
Match for Pattern 1 found in 5F0L_3|Chain C|Sorting nexin-3|Homo sapiens (9606) at position 70
Match for Pattern 1 found in 5F0M_3|Chain C|Sorting nexin-3|Homo sapiens (9606) at position 70
Match for Pattern 1 found in 5F0P_3|Chain C|Sorting nexin-3|Homo sapiens (9606) at position 70
Match for Pattern 2 found in 5JPM_2|Chains B,E|Complement C4-A|Homo sapiens (9606) at position 252
Match for Pattern 2 found in 5JPN_2|Chain B|Complement C4-A|Homo sapiens (9606) at position 252
Match for Pattern 2 found in 5JTW_2|Chains B,E|Complement C4-A|Homo sapiens (9606) at position 175
Match for Pattern 1 found in 5N69_1|Chains A,B|Myosin-7|Bos taurus (9913) at position 69
Match for Pattern 1 found in 5N6A_1|Chain A|Myosin-7|Bos taurus (9913) at position 69
Match for Pattern 1 found in 5TBY_1|Chains A,B|Myosin-7|Homo sapiens (9606) at position 69
Match for Pattern 1 found in 6FSA_1|Chains A,B|Myosin-7|Bos taurus (9913) at position 69
Match for Pattern 1 found in 6XSZ_3|Chains D,G,J|Myosin-7|Bos taurus (9913) at position 69
Match for Pattern 2 found in 6YSQ_1|Chains A,B|Complement C4-B, Complement C4-B|Homo sapiens (9606) at position 854
Match for Pattern 1 found in 7JH7_2|Chains F,G,H|Myosin-7|Sus scrofa (9823) at position 69
```

In []:

Question 6

UniProtKB 968 results

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
P66948	BEPA_ECOLI	Beta-barrel assembly-enhancing protease[...]	bepA, yfgC, b2494, JW2479	Escherichia coli (strain K12)	487 AA
P0A910	OMPA_ECOLI	Outer membrane protein A[...]	ompA, con, tolQ, tut, b0957, JW0940	Escherichia coli (strain K12)	346 AA
Q1B8U1	MONAL_PSEE4	Monalysin[...]	mnt, PSEEN3174	Pseudomonas entomophila (strain L48)	271 AA
O18423	TXL_EISFE	Lysexin[...]		Eisenia fetida (Red wiggler worm)	297 AA
P0A937	BAM_ECOLI	Outer membrane protein assembly factor BamE	bamE, smpA, b2617, JW2598	Escherichia coli (strain K12)	113 AA
P0AC02	BAMD_ECOLI	Outer membrane protein assembly factor BamD	bamD, yf0, b2595, JW2577	Escherichia coli (strain K12)	245 AA
P06996	OMPC_ECOLI	Outer membrane porin C[...]	ompC, meoA, par, b2215, JW2203	Escherichia coli (strain K12)	367 AA
P77774	BAMB_ECOLI	Outer membrane protein assembly factor BamB	bamB, yfgL, b2512, JW2496	Escherichia coli (strain K12)	392 AA
P62072	TIM10_HUMAN	Mitochondrial import inner membrane translocase subunit Tim10	TIMM10, TIM10	Homo sapiens (Human)	90 AA
P0A903	BAMC_ECOLI	Outer membrane protein assembly factor BamC	bamC, dapX, nlpB, b2477, JW2462	Escherichia coli (strain K12)	344 AA
Q9YSJ4	TIM13_HUMAN	Mitochondrial import inner membrane translocase subunit Tim13	TIMM13, TIM13B, TIMM13A, TIMM13B	Homo sapiens (Human)	95 AA
Q9YSJ7	TIM9_HUMAN	Mitochondrial import inner membrane translocase subunit Tim9	TIMM9, TIM9, TIM9A, TIMM9A	Homo sapiens (Human)	85 AA
P0A940	BAMA_ECOLI	Outer membrane protein assembly factor BamA[...]	bamA, yaeT, yzzX, yzzY, b0177, JW0172	Escherichia coli (strain K12)	810 AA
A0A2S4N3N0	OMPA_SHIFL	Outer membrane protein A[...]	ompA, SF0957	Shigella flexneri	348 AA
P05825	FEPA_ECOLI	Ferrienterobactin receptor[...]	fepA, fep, feuB, b0584, JW5086	Escherichia coli (strain K12)	746 AA
P14683	SAM35_YEAST	Sorting assembly machinery 35 kDa subunit[...]	SAM35, OMP85, TOB30, TOM30, YHR083W	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	329 AA
Q9Y277	VDAC3_HUMAN	Voltage dependent anion-selective channel protein 3[...]	VDAC3	Homo sapiens (Human)	283 AA

the Sequence Manipulation Suite is used to identify the given pattern in the FASTA file downloaded. The pattern is modified to [KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW] to work well with the server. 466 hits were found.

Sequence Manipulation Suite - Google Chrome

about:blank

Protein Pattern Find results

Results for 348 residue sequence "sp|A0A2S4N3N0|OMPA_SHIFL Outer membrane protein A OS=Shigella flexneri OX=623 GN=ompA PE=1 SV=1" starting "MKKT/ no matches found for this sequence.

Results for 713 residue sequence "sp|A1L314|MPEGI_MOUSE Macrophage-expressed gene 1 protein OS=Mus musculus OX=10090 GN=Mpeg1 PE=1 SV=1" starting "RAGLPLHF

>match number 1 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=305 end=312

>match number 2 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=516 end=523

Results for 1350 residue sequence "sp|A1Z877|NDG_DROME Nidogen OS=Drosophila melanogaster OX=7227 GN=Ndg PE=1 SV=1" starting "MPTFGSKLLA"

>match number 1 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=369 end=376

>match number 2 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=583 end=590

Results for 5100 residue sequence "sp|A2AJ76|HMCN2_MOUSE Hemicentin-2 OS=Mus musculus OX=10090 GN=Hmcn2 PE=1 SV=1" starting "MTPGAQLLFL"

>match number 1 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=734 end=741

>match number 2 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=2002 end=2009

>match number 3 to "[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]" start=3383 end=3390

466 hits were found as shown below



ExPASy <expasy@expasy.org>
to me ▾

11:56 PM (0 minutes ago) ☆ ↶ ⋮

ScanProsite results:

Output format: Text

Hits for USERPAT1 "[KRHQFE]-x-G-[VLFAC]-x-[VLFMYYW]-x-[VLFW]" on UyxXAgOI9 custom database sequences:

found: 466 hits in 321 sequences

Graphical View (graphical view with feature detection): [https://p-prosite.expasy.org/cgi-bin/prosite/ScanView.cgi?scanfile=443236654235_scan.gz&sig=\[KRHQFE\]-x-G-\[VLFAC\]-x-\[VLFMYYW\]-x-\[VLFW\]](https://p-prosite.expasy.org/cgi-bin/prosite/ScanView.cgi?scanfile=443236654235_scan.gz&sig=[KRHQFE]-x-G-[VLFAC]-x-[VLFMYYW]-x-[VLFW])
(link will be valid for 12hours)

Hits for:

>USERPAT1 (user pattern):

Pattern: [KRHQFE]-x-G-[VLFAC]-x-[VLFMYYW]-x-[VLFW]

Approximate number of expected random matches in ~ 100'000 sequences (50'000'000 residues): 26784

>sp|Q06305|AER3_AERHY (492 aa)

Aerolysin-3 (Hemolysin-3). [Aeromonas hydrophila]

MKKLKITGLSLIISGLLMAQAQAAEPVYPDQLRLFSLGQEVCGDKYRPVNREEAQSVKSNIVGMMGQWQISGLANGWVIM

GPGYNGEIKPGSASSTWCYPTNPATGEIPTLSALDIIPDGDEVQVQWRLVHDSANFIKPTSYLAHYLG YAWVGGNHSQYVG